IGNOU BCS-012: Basic Mathematics

Term-End Examination, December 2023 - Full Solved Paper

Time: 3 Hours Maximum Marks: 100

How to use this solved paper

Question 1 is compulsory. In the actual examination, any three questions are attempted from Questions 2–5. Here, all questions have been solved for complete practice. A compact theory and formula sheet is given at the end.

Question 1

1(a). Show that

$$\begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix} = 0.$$

Solution. Add the elements of the first row:

$$(b-c)+(c-a)+(a-b)=0.$$

Similarly, the sum of every row is zero. Therefore, if the three columns are added, we get a zero column:

$$C_1+C_2+C_3=0.$$

Thus the columns are linearly dependent. Hence the determinant is zero.

$$\boxed{0}$$

1(b). If

$$A=\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}, \qquad f(x)=x^2-4x+7,$$

show that $f(A)=O_{2\times 2}$. Use this result to find A^5.

Solution. First,

$$A^2=\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}=\begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix}.$$

Hence

$$f(A)=A^2-4A+7I=\begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix}-\begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix}+\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus

$$A^2-4A+7I=0 \quad\Rightarrow\quad A^2=4A-7I.$$

Now,

$$A^3=A(4A-7I)=4A^2-7A=4(4A-7I)-7A=9A-28I,$$
$$A^4=A(9A-28I)=9A^2-28A=9(4A-7I)-28A=8A-63I,$$
$$A^5=A(8A-63I)=8A^2-63A=8(4A-7I)-63A=-31A-56I.$$

Therefore,

$$A^5=-31\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}-56\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}.$$

$$\boxed{A^5 = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}}$$

1(c). Show that 7 divides $2^{3n} - 1$ for every $n \in \mathbb{N}$.

Solution. Since $2^3 = 8$, we have

$$2^{3n} - 1 = 8^n - 1.$$

Now $8 \equiv 1 \pmod 7$. Therefore,

$$8^n \equiv 1^n \equiv 1 \pmod 7.$$

Thus

$$8^n - 1 \equiv 0 \pmod 7.$$

Hence 7 divides $2^{3n} - 1$ for every natural number n.

$$\boxed{7 \mid (2^{3n} - 1)}$$

1(d). If $1, \omega, \omega^2$ are cube roots of unity, show that

$$(1+\omega)(1+\omega^2)(1+\omega^3)(1+\omega^4)(1+\omega^6)(1+\omega^8) = 4.$$

Solution. For cube roots of unity,

$$\omega^3 = 1, \qquad 1 + \omega + \omega^2 = 0.$$

Reduce powers modulo 3:

$$\omega^3 = 1, \quad \omega^4 = \omega, \quad \omega^6 = 1, \quad \omega^8 = \omega^2.$$

Therefore,

$$(1+\omega)(1+\omega^2)(1+\omega^3)(1+\omega^4)(1+\omega^6)(1+\omega^8)$$
$$= (1+\omega)(1+\omega^2)(2)(1+\omega)(2)(1+\omega^2).$$

Also,

$$(1+\omega)(1+\omega^2) = 1 + \omega + \omega^2 + \omega^3 = 1 + (-1) + 1 = 1.$$

Hence the product is

$$4\{(1+\omega)(1+\omega^2)\}^2 = 4.$$

$$\boxed{4}$$

1(e). If $y = ae^{mx} + be^{-mx} + 4$, show that

$$\frac{d^2y}{dx^2} = m^2(y - 4).$$

Solution. Given

$$y = ae^{mx} + be^{-mx} + 4.$$

Then

$$\frac{dy}{dx} = ame^{mx} - bme^{-mx},$$

$$\frac{d^2y}{dx^2} = am^2e^{mx} + bm^2e^{-mx} = m^2(ae^{mx} + be^{-mx}).$$

But

$$y - 4 = ae^{mx} + be^{-mx}.$$

Therefore,

$$\frac{d^2y}{dx^2} = m^2(y-4).$$

$$\boxed{\frac{d^2y}{dx^2} = m^2(y-4)}$$

1(f). If α, β are roots of

$$x^2 - 2kx + k^2 - 1 = 0$$

and $\alpha^2 + \beta^2 = 10$, find k.

Solution. For the quadratic equation,

$$\alpha + \beta = 2k, \qquad \alpha\beta = k^2 - 1.$$

Now

$$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta.$$

Thus

$$10 = (2k)^2 - 2(k^2-1) = 4k^2 - 2k^2 + 2 = 2k^2 + 2.$$

So

$$2k^2 = 8 \quad \Rightarrow \quad k^2 = 4.$$

$$\boxed{k = \pm 2}$$

1(g). Find the value of λ for which the vectors

$$\vec{a} = 2\hat{i} - 4\hat{j} + 3\hat{k}, \quad \vec{b} = \lambda\hat{i} - 2\hat{j} + \hat{k}, \quad \vec{c} = 2\hat{i} + 3\hat{j} + 3\hat{k}$$

are coplanar.

Solution. Three vectors are coplanar if their scalar triple product is zero:

$$[\vec{a}\ \vec{b}\ \vec{c}] = 0.$$

Thus

$$\begin{vmatrix} 2 & -4 & 3 \\ \lambda & -2 & 1 \\ 2 & 3 & 3 \end{vmatrix} = 0.$$

Expanding,

$$2\begin{vmatrix} -2 & 1 \\ 3 & 3 \end{vmatrix} + 4\begin{vmatrix} \lambda & 1 \\ 2 & 3 \end{vmatrix} + 3\begin{vmatrix} \lambda & -2 \\ 2 & 3 \end{vmatrix} = 0.$$

$$2(-9) + 4(3\lambda - 2) + 3(3\lambda + 4) = 0.$$

$$-18 + 12\lambda - 8 + 9\lambda + 12 = 0 \quad \Rightarrow \quad 21\lambda - 14 = 0.$$

$$\boxed{\lambda = \frac{2}{3}}$$

1(h). Find the angle between the pair of lines

$$\frac{x-5}{2} = \frac{y-3}{3} = \frac{z-1}{-3}$$

and

$$\frac{x}{3} = \frac{y-1}{2} = \frac{z+5}{-3}.$$

Solution. The direction ratios of the two lines are

$$\vec{d_1} = (2, 3, -3), \qquad \vec{d_2} = (3, 2, -3).$$

The angle θ between the lines is the angle between their direction vectors:

$$\cos\theta = \frac{|\vec{d_1} \cdot \vec{d_2}|}{|\vec{d_1}||\vec{d_2}|}.$$

Now

$$\vec{d_1} \cdot \vec{d_2} = 2 \cdot 3 + 3 \cdot 2 + (-3)(-3) = 21,$$
$$|\vec{d_1}| = \sqrt{4+9+9} = \sqrt{22}, \quad |\vec{d_2}| = \sqrt{9+4+9} = \sqrt{22}.$$

Therefore,

$$\cos\theta = \frac{21}{22}.$$

$$\boxed{\theta = \cos^{-1}\left(\frac{21}{22}\right)}$$

Question 2

2(a). Solve the following set of linear equations by using matrix inverse:

$$3x + 4y + 7z = -2, \quad 2x - y + 3z = 6, \quad 2x + 2y - 3z = 0.$$

Solution. Write the system as $AX = B$, where

$$A = \begin{bmatrix} 3 & 4 & 7 \\ 2 & -1 & 3 \\ 2 & 2 & -3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} -2 \\ 6 \\ 0 \end{bmatrix}.$$

The inverse method gives

$$X = A^{-1}B.$$

Solving, we obtain

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}.$$

Verification:

$$3(2) + 4(-2) + 7(0) = -2,$$
$$2(2) - (-2) + 3(0) = 6,$$
$$2(2) + 2(-2) - 3(0) = 0.$$

$$\boxed{x = 2, \quad y = -2, \quad z = 0}$$

2(b). Use the principle of mathematical induction to prove that

$$1^3 + 2^3 + \cdots + n^3 = \frac{1}{4}n^2(n+1)^2$$

for every natural number n.

Solution. Let

$$P(n) : 1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

For $n = 1$,

$$1^3 = 1 = \frac{1^2(1+1)^2}{4}.$$

So $P(1)$ is true.

Assume $P(k)$ is true:

$$1^3 + 2^3 + \cdots + k^3 = \frac{k^2(k+1)^2}{4}.$$

Then

$$1^3 + 2^3 + \cdots + k^3 + (k+1)^3 = \frac{k^2(k+1)^2}{4} + (k+1)^3.$$

$$= (k+1)^2 \left(\frac{k^2}{4} + k + 1\right) = (k+1)^2 \left(\frac{k^2 + 4k + 4}{4}\right).$$

$$= \frac{(k+1)^2(k+2)^2}{4}.$$

This is the required formula for $n = k + 1$. Therefore, by induction, the result holds for every $n \in \mathbb{N}$.

$$\boxed{1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2}$$

2(c). Find how many terms of the GP

$$\sqrt{3},\ 3,\ 3\sqrt{3}, \ldots$$

add up to $120 + 40\sqrt{3}$.

Solution. Here

$$a = \sqrt{3}, \qquad r = \frac{3}{\sqrt{3}} = \sqrt{3}.$$

The sum of n terms is

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{\sqrt{3}\{(\sqrt{3})^n - 1\}}{\sqrt{3} - 1}.$$

Check the sum of the first eight terms:

$$\sqrt{3} + 3 + 3\sqrt{3} + 9 + 9\sqrt{3} + 27 + 27\sqrt{3} + 81.$$

Separate rational and irrational parts:

$$(3 + 9 + 27 + 81) + (1 + 3 + 9 + 27)\sqrt{3} = 120 + 40\sqrt{3}.$$

Hence the required number of terms is eight.

$$\boxed{n = 8}$$

2(d). Write De Moivre's theorem and use it to find $(i + \sqrt{3})^3$.

Solution. De Moivre's theorem states that if n is an integer, then

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

Now

$$i + \sqrt{3} = \sqrt{3} + i.$$

Its modulus is

$$r = \sqrt{(\sqrt{3})^2 + 1^2} = 2.$$

Also

$$\cos\theta = \frac{\sqrt{3}}{2}, \qquad \sin\theta = \frac{1}{2},$$

so $\theta = \pi/6$. Therefore,

$$\sqrt{3} + i = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right).$$

Thus

$$(i+\sqrt{3})^3 = 2^3\left(\cos\frac{3\pi}{6} + i\sin\frac{3\pi}{6}\right) = 8\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 8i.$$

$$\boxed{(i+\sqrt{3})^3 = 8i}$$

Question 3

3(a). If

$$A = \begin{bmatrix} -1 & 2 & 3 \\ 4 & 5 & 7 \\ 5 & 3 & 4 \end{bmatrix},$$

show that $A(\text{adj}\,A) = 0$.

Solution. We know that

$$A(\text{adj}\,A) = |A|I.$$

So it is enough to show that $|A| = 0$.

$$|A| = \begin{vmatrix} -1 & 2 & 3 \\ 4 & 5 & 7 \\ 5 & 3 & 4 \end{vmatrix}.$$

Expanding along the first row,

$$|A| = -1\begin{vmatrix} 5 & 7 \\ 3 & 4 \end{vmatrix} - 2\begin{vmatrix} 4 & 7 \\ 5 & 4 \end{vmatrix} + 3\begin{vmatrix} 4 & 5 \\ 5 & 3 \end{vmatrix}.$$

$$= -1(20-21) - 2(16-35) + 3(12-25) = 1 + 38 - 39 = 0.$$

Therefore,

$$A(\text{adj}\,A) = |A|I = 0I = 0.$$

$$\boxed{A(\text{adj}\,A) = 0}$$

3(b). Solve the inequality

$$\left|\frac{x-4}{2}\right| \le \frac{5}{12}$$

and graph the solution set.

Solution.

$$\left|\frac{x-4}{2}\right| \le \frac{5}{12} \quad \Rightarrow \quad |x-4| \le \frac{5}{6}.$$

Thus

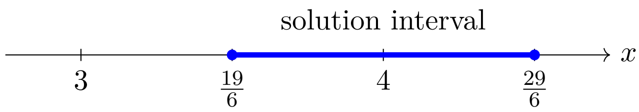
$$-\frac{5}{6} \le x - 4 \le \frac{5}{6}.$$

Adding 4 throughout,

$$4 - \frac{5}{6} \le x \le 4 + \frac{5}{6}.$$

Therefore,

$$\frac{19}{6} \le x \le \frac{29}{6}.$$



$$\boxed{x \in \left[\frac{19}{6}, \frac{29}{6}\right]}$$

3(c). Solve the equation

$$8x^3 - 14x^2 + 7x - 1 = 0,$$

given that the roots are in GP.

Solution. We factor the polynomial:

$$8x^3 - 14x^2 + 7x - 1 = (x-1)(2x-1)(4x-1).$$

Therefore,

$$(x-1)(2x-1)(4x-1) = 0.$$

So

$$x = 1, \quad x = \frac{1}{2}, \quad x = \frac{1}{4}.$$

These roots are in GP because

$$\frac{1/2}{1/4} = 2, \qquad \frac{1}{1/2} = 2.$$

$$\boxed{x = \frac{1}{4}, \frac{1}{2}, 1}$$

3(d). Verify that

$$f(x) = 1 + x^2 \ln\left(\frac{1}{x}\right)$$

has a local maxima at $x = \frac{1}{\sqrt{e}}$, $x > 0$.

Solution. Since

$$\ln\left(\frac{1}{x}\right) = -\ln x,$$

we write

$$f(x) = 1 - x^2 \ln x.$$

Then

$$f'(x) = -(2x \ln x + x) = -x(2\ln x + 1).$$

For a stationary point,

$$f'(x) = 0.$$

Since $x > 0$,

$$2\ln x + 1 = 0 \quad \Rightarrow \quad \ln x = -\frac{1}{2} \quad \Rightarrow \quad x = e^{-1/2} = \frac{1}{\sqrt{e}}.$$

Now

$$f''(x) = -(2\ln x + 3).$$

At $x = 1/\sqrt{e}$,

$$\ln x = -\frac{1}{2}, \quad f''(x) = -\{-1 + 3\} = -2 < 0.$$

Hence $f(x)$ has a local maximum at $x = 1/\sqrt{e}$. The maximum value is

$$f\left(\frac{1}{\sqrt{e}}\right) = 1 + \frac{1}{e} \cdot \frac{1}{2} = 1 + \frac{1}{2e}.$$

$$\boxed{\text{Local maximum at } x = \frac{1}{\sqrt{e}}, \quad f_{\max} = 1 + \frac{1}{2e}}$$

Question 4

4(a). Evaluate

$$\lim_{x \to 0} \frac{\sqrt{1+2x} - \sqrt{1-2x}}{x}.$$

Solution. Rationalise the numerator:

$$\frac{\sqrt{1+2x} - \sqrt{1-2x}}{x} \cdot \frac{\sqrt{1+2x} + \sqrt{1-2x}}{\sqrt{1+2x} + \sqrt{1-2x}}.$$

This gives

$$\frac{(1+2x) - (1-2x)}{x\{\sqrt{1+2x} + \sqrt{1-2x}\}} = \frac{4x}{x\{\sqrt{1+2x} + \sqrt{1-2x}\}}.$$

So

$$\lim_{x \to 0} \frac{4}{\sqrt{1+2x} + \sqrt{1-2x}} = \frac{4}{1+1} = 2.$$

$$\boxed{2}$$

4(b). Find the shortest distance between the lines

$$\vec{r}_1 = (3\hat{i} + 4\hat{j} - 2\hat{k}) + t(-\hat{i} + 2\hat{j} + \hat{k})$$

and

$$\vec{r}_2 = (\hat{i} - 7\hat{j} - 2\hat{k}) + t(\hat{i} + 3\hat{j} + 2\hat{k}).$$

Solution. Let

$$\vec{a}_1 = (3, 4, -2), \quad \vec{b}_1 = (-1, 2, 1),$$
$$\vec{a}_2 = (1, -7, -2), \quad \vec{b}_2 = (1, 3, 2).$$

The shortest distance between skew lines is

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}.$$

Now

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 1 \\ 1 & 3 & 2 \end{vmatrix} = \hat{i} + 3\hat{j} - 5\hat{k}.$$

Also,

$$\vec{a}_2 - \vec{a}_1 = (1, -7, -2) - (3, 4, -2) = (-2, -11, 0).$$

Therefore,

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = (-2)(1) + (-11)(3) + 0(-5) = -35.$$
$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{1^2 + 3^2 + (-5)^2} = \sqrt{35}.$$

Hence

$$d = \frac{35}{\sqrt{35}} = \sqrt{35}.$$

$$\boxed{\sqrt{35}}$$

4(c). Determine the length of the curve

$$y = \frac{2}{3}x^{3/2}$$

from $(0, 0)$ **to** $(1, \frac{2}{3})$**.**

Solution. The arc length is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

Given

$$y = \frac{2}{3}x^{3/2}, \quad \frac{dy}{dx} = \frac{2}{3} \cdot \frac{3}{2}x^{1/2} = \sqrt{x}.$$

From $x = 0$ to $x = 1$,

$$L = \int_0^1 \sqrt{1+x}\, dx.$$

Now

$$\int \sqrt{1+x}\, dx = \frac{2}{3}(1+x)^{3/2}.$$

Therefore,

$$L = \left[\frac{2}{3}(1+x)^{3/2}\right]_0^1 = \frac{2}{3}\left(2^{3/2} - 1\right).$$

$$\boxed{L = \frac{2}{3}(2\sqrt{2} - 1)}$$

4(d). Find the sum of all integers between 100 **and** 1000 **that are divisible by** 7**.**

Solution. The first integer greater than 100 divisible by 7 is

$$105 = 7 \times 15.$$

The last integer less than 1000 divisible by 7 is

$$994 = 7 \times 142.$$

Thus the required numbers form an AP:

$$105, 112, 119, \ldots, 994.$$

The number of terms is

$$n = \frac{994 - 105}{7} + 1 = 127 + 1 = 128.$$

Therefore,

$$S_n = \frac{n}{2}(a + l) = \frac{128}{2}(105 + 994) = 64 \times 1099 = 70336.$$

$$\boxed{70336}$$

Question 5

5(a). Determine the area between the two curves

$$y = 3 + 2x, \qquad y = 3 - x, \quad 0 \le x \le 3,$$

using integration.

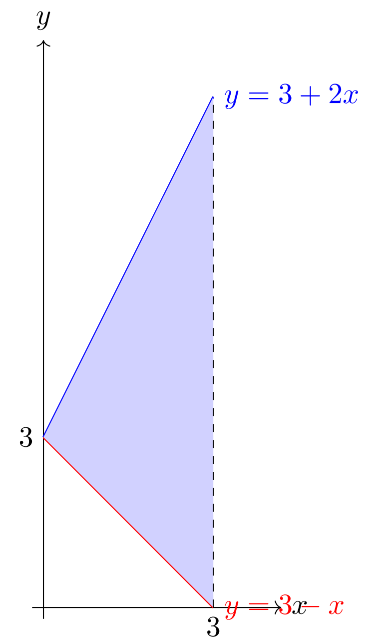
Solution. For $0 \le x \le 3$,

$$(3 + 2x) - (3 - x) = 3x \ge 0.$$

So $y = 3 + 2x$ is the upper curve and $y = 3 - x$ is the lower curve. The required area is

$$A = \int_0^3 \{(3 + 2x) - (3 - x)\}\, dx = \int_0^3 3x\, dx.$$

$$A = \left[\frac{3x^2}{2}\right]_0^3 = \frac{27}{2}.$$



$$\boxed{\text{Area} = \frac{27}{2} \text{ square units}}$$

5(b). Find the direction cosines of the line passing through the points $(1, 2, 3)$ and $(-1, 1, 0)$.

Solution. The direction vector of the line is

$$\vec{d} = (-1 - 1,\ 1 - 2,\ 0 - 3) = (-2, -1, -3).$$

Its magnitude is

$$|\vec{d}| = \sqrt{(-2)^2 + (-1)^2 + (-3)^2} = \sqrt{14}.$$

Therefore, the direction cosines are

$$l = \frac{-2}{\sqrt{14}}, \quad m = \frac{-1}{\sqrt{14}}, \quad n = \frac{-3}{\sqrt{14}}.$$

If the opposite direction is chosen, all signs may be changed simultaneously.

$$\boxed{\left(-\frac{2}{\sqrt{14}},-\frac{1}{\sqrt{14}},-\frac{3}{\sqrt{14}}\right)}$$

5(c). Find the maximum value of $2a + 5b$ subject to

$$-3a-2b \le -6, \quad -2a+b \le 2, \quad 4a+6b \le 24, \quad 2a-3b \le 3, \quad a \ge 0, \quad b \ge 0.$$

Solution. First rewrite the constraints:

$$3a+2b \ge 6, \quad -2a+b \le 2, \quad 2a+3b \le 12, \quad 2a-3b \le 3, \quad a,b \ge 0.$$

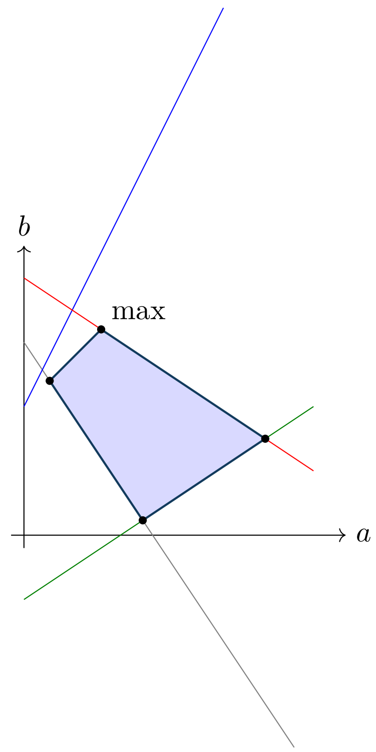
The corner points of the feasible region are obtained from intersections of boundary lines. The relevant vertices are

$$\left(\frac{2}{5},\frac{12}{5}\right), \quad \left(\frac{6}{5},\frac{16}{5}\right), \quad \left(\frac{24}{13},\frac{3}{13}\right), \quad \left(\frac{15}{4},\frac{3}{2}\right).$$

Evaluate $Z = 2a + 5b$ at these vertices:

(a,b)	$Z = 2a+5b$
$\left(\frac{2}{5},\frac{12}{5}\right)$	$\frac{64}{5}$
$\left(\frac{6}{5},\frac{16}{5}\right)$	$\frac{92}{5}$
$\left(\frac{24}{13},\frac{3}{13}\right)$	$\frac{63}{13}$
$\left(\frac{15}{4},\frac{3}{2}\right)$	15

The maximum is $\frac{92}{5}$ at $\left(\frac{6}{5},\frac{16}{5}\right)$.



$$\boxed{Z_{\max} = \frac{92}{5} \text{ at } (a,b) = \left(\frac{6}{5},\frac{16}{5}\right)}$$

5(d). Reduce the matrix

$$A = \begin{bmatrix} 5 & 3 & 8 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

to normal form and hence find its rank.

Solution. Use elementary row operations:

$$A = \begin{bmatrix} 5 & 3 & 8 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Interchange R_1 and R_3:

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 5 & 3 & 8 \end{bmatrix}.$$

Apply $R_3 \to R_3 - 5R_1$:

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 8 & 8 \end{bmatrix}.$$

Apply $R_3 \to R_3 - 8R_2$:

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Apply $R_1 \to R_1 + R_2$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

By using elementary column operations, subtract column 1 from column 3, then subtract column 2 from column 3:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

This is the normal form.

$$\boxed{\text{Normal form} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(A) = 2}$$

Theory and Formula Sheet Required for This Paper

1. Determinants and matrices

- If one row or column is a linear combination of others, the determinant is zero.
- $A(\text{adj}\, A) = |A|I$.
- If $f(A) = 0$, use it to reduce higher powers of A.
- A matrix is in normal form $\text{diag}(I_r, 0)$, where r is its rank.

2. Algebra, sequences, induction

- For $ax^2 + bx + c = 0$, sum of roots $= -b/a$, product of roots $= c/a$.
- GP sum: $S_n = \dfrac{a(r^n - 1)}{r - 1}$, $r \neq 1$.
- Sum of cubes: $1^3 + 2^3 + \cdots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2$.
- Mathematical induction: prove base case, assume $P(k)$, then prove $P(k+1)$.

3. Complex numbers and cube roots of unity

- Cube roots of unity are $1, \omega, \omega^2$ with $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$.
- De Moivre's theorem: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
- Write $a + ib = r(\cos\theta + i\sin\theta)$ before applying De Moivre's theorem.

4. Calculus

- For local maxima/minima, solve $f'(x) = 0$ and use $f''(x)$: if $f'' < 0$, local maximum; if $f'' > 0$, local minimum.
- Arc length: $L = \int_a^b \sqrt{1 + (dy/dx)^2}\, dx$.
- Area between curves: $A = \int_a^b (\text{upper curve} - \text{lower curve})\, dx$.
- Rationalisation is useful for limits involving square roots.

5. Three-dimensional geometry and vectors

- Direction cosines of vector (a, b, c) are $\left(\frac{a}{r}, \frac{b}{r}, \frac{c}{r}\right)$, where $r = \sqrt{a^2 + b^2 + c^2}$.
- Three vectors are coplanar if their scalar triple product is zero.
- Angle between lines with direction vectors $\vec{a}, \vec{b}$:

$$\cos\theta = \frac{|\vec{a} \cdot \vec{b}|}{|\vec{a}||\vec{b}|}.$$

- Shortest distance between skew lines:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}.$$

6. Linear programming

- Convert inequalities to boundary lines and find the feasible region.
- The maximum or minimum of a linear objective function occurs at a corner point of the feasible region.
- Evaluate the objective function at every feasible vertex and choose the best value.